

GRADE 6 MATHEMATICS

Classroom Observation Guide

Grade 6 Mathematics — Day Five: "Area Is Multiplication"

Professor Xavier Honablue, M.Ed.

Prepared for visiting guests, administrators, and instructional observers.

Purpose of This Guide

Thank you for observing our Grade 6 mathematics classroom today. Days One through Four built the routines: graph-paper construction, the class data wall, skits and votes, and the Engage/Explore/Enrich station model. Today is the first day of grade-level content. The district curriculum, i-Ready Classroom Mathematics, opens Grade 6 with Unit 1, Lesson 1: Find the Area of a Parallelogram. This guide explains how that lesson has been adapted for this class, the Common Core and Michigan standards it addresses, and the research behind the instructional choices you will see.

The Design Decision: Teach Area, Repair Multiplication Inside It

Diagnostic work during Days One through Four showed that most students cannot yet multiply two-digit numbers, and roughly half are not fluent with single-digit multiplication facts. The published i-Ready session assumes rectangle area is automatic and moves straight to parallelograms. The instructional question was whether to pause for a multiplication unit or proceed with area.

The decision was to do both at once. On grid paper, the area of a rectangle is the array model of multiplication, and the “box method” for two-digit multiplication is simply a rectangle cut into tens and ones. Students therefore spend today on rectangles, split each side at ten, find four small pieces using only single-digit facts, and add. They learn two-digit multiplication as a by-product of doing grade-level geometry, rather than being pulled off grade-level work to remediate.

A Second Diagnosis: Spelling and Handwriting

Days One through Four also surfaced a second, quieter gap alongside the multiplication gap: much of this class struggles with spelling and handwriting. Students who can reason about a rectangle out loud often stall the moment they have to write the reasoning down — a missed letter, a reversed digit, or a name that takes three tries to print legibly. Left unaddressed, that gap would make every written task in this unit measure handwriting and spelling as much as it measures mathematics.

The response mirrors the one used for multiplication: repair the skill inside the grade-level task rather than pulling students aside to work on it separately. Today's classwork worksheet, described below, keeps the writing load small and structured — letter boxes instead of blank lines for a name, a fill-in-the-blank vocabulary word instead of an open-ended definition, and sentence frames with a word bank instead of unscaffolded sentences — so that a shaky hand or an uncertain speller is not shut out of the mathematics.

$$14 \times 23 = 200 + 30 + 80 + 12 = 322 \text{ square units}$$

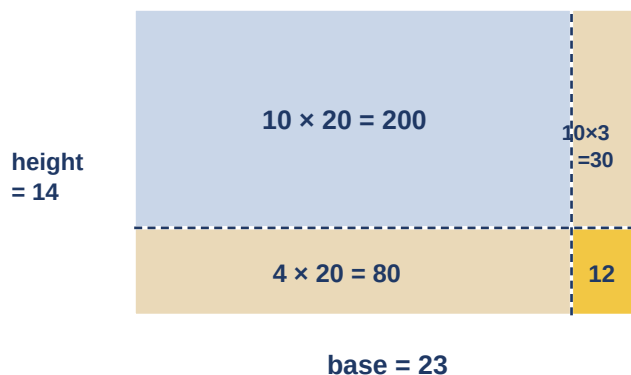


Figure 1. The teacher's model rectangle. Every piece uses a single-digit fact and a ten.

Lesson at a Glance (55 Minutes)

Time	Segment	What's Happening
0-3 min	Welcome Back	Students sign in on the lesson app (name, shape avatar, learning mode) and hear the goal: by the end of class you will multiply 14×23 by drawing.
3-8 min	Facts Sprint	Ten single-digit facts answered in journals and checked on the app. Students stand behind their score on the Human Bar Graph line. This becomes a standing 5-minute daily routine.
8-13 min	Try It (Engage)	A 6×8 rectangle on a grid. Students find the number of squares a slow way and a fast way, then name the idea: area = rows $\times$ columns. $A = b \times h$ is written on the board.
13-28 min	Explore: Target Dimensions	Teams throw two Nerf darts at the scoring board; dart 1 is the base, dart 2 the height (bullseye = 25). Teams draw the rectangle on grid paper, split each side at 10, find four pieces, and add. Goal: three rectangles per team.
28-38 min	Discuss It / Connect It	Two teams present under the document camera. The class locates each number of the expression on the picture. The teacher writes $(10 + 4)(20 + 3)$ as four partial products and substitutes into $A = b \times h$. A short skit ("Counter vs. Splitter") ends in a class vote.

Time	Segment	What's Happening
38-48 min	Engage / Explore / Enrich	Three boards, assigned by what the teacher observed during Explore. Engage builds arrays with the teacher; Explore works the four-piece split independently; Enrich cuts a parallelogram and slides it into a rectangle (the actual i-Ready session content) as a preview of Day Six. Each board ends on an aligned IXL skill.
48-53 min	Apply It: Exit Ticket	Three problems, silent, on paper and app: 7×9 ; 24×13 with the split; area 96 and base 8, find the height. Enrich adds a parallelogram problem.
53-55 min	Journal and Keep-or-Rid	Journal: draw a rectangle that shows 12×15 and mark where you would cut it. Class votes whether to keep the dart board for Day Six; the tally is recorded on the data wall.

Why a Nerf Dart Board?

A standard dart board scores 1 through 20 with a 25-point bullseye. That range means every throw produces a side length small enough to draw on grid paper and split at ten, and large enough to require two-digit multiplication. The board is a random dimension generator with a reason to care about the result: the team owns the rectangle it threw. Darts move only teacher-to-thrower-to-teacher; a miss earns one re-throw, then the team draws a number card. If the protocol breaks, the card deck replaces the board for the rest of the period.

Today's Classwork Worksheet: "Multiplication Means Growth"



Section A, Guess the Word: two missing letters complete GROWTH.

Each student receives a one-page classwork worksheet, Multiplication Means Growth, at the start of class (CCSS 3.OA.A.1, 6.G.A.1, and the ELA standard L.6.4 — determining the meaning of a word from context). It opens with Section A, Guess the Word: students fill in two missing letters to spell GROWTH, then complete the sentence "Multiplication means ____." Section B, Meet the Multiplication Family, uses a word bank (number of groups, size of each group, product, growth) to define the two factors and the product in a multiplication expression. Section C, Try It: 4×3 , asks students to label a worked example and then draw their own array (5 groups of 2) before a short Explore/Summary panel previews today's big idea — that the same groups from Section C, lined up into rows and columns, become the rectangle and area model used in Target Dimensions.

Rather than treating vocabulary as separate from the day's mathematics, the worksheet is built so that spelling the word and modeling the idea are the same act: a student who cannot yet write "growth" confidently can still show what it means by drawing the dots and the rectangle, and a student who draws the model correctly has independent evidence for what the word they just spelled describes.

The Visualizing-Simulation Skill in the ELA + Math Literacy Approach

This worksheet is one piece of a broader ELA + Math literacy approach used throughout the unit: every new mathematical term is first met as a word to construct, letter by letter, and then immediately met a second time as something to see. In today's lesson, that second encounter is a simulation — the Target Dimensions dart board. A team's two dart throws generate a base and a height; drawing that rectangle and splitting it at ten turns an abstract pair of numbers into a picture the student built with their own throw. The word GROWTH from Section A and the rectangle from Target Dimensions describe the same idea — a small number multiplying into a much larger total — so students who struggle to hold the definition in words still have the simulation to fall back on, and students who struggle with the drawing still have the word.

This pairing of a verbal/orthographic representation with a visual, simulated one draws on dual coding theory (Paivio, 1971) and Mayer's (2009) research on multimedia learning, both of which find that information encoded through two channels — words and images — is retained and understood better than either channel alone, particularly for students who are still building fluency in one of the two.

The Engage / Explore / Enrich Boards

Board	Who	Task	IXL skill
Engage (with teacher)	Could not point to 10×3 on the picture, or sprint under 6/10	Build 3×7 , 4×6 , 6×8 as arrays, counting rows aloud; then 12×6 split at ten only. Names go on the week's 1-on-1 tutoring list.	CQZ – Tile a rectangle and find the area
Explore (independent)	Got the split with support; facts mostly there	Three rectangles from the card deck with the four-piece split; one backwards problem (area 96, base 8).	8PN – Area of rectangles and squares
Enrich (independent)	Fluent on the split, finished early	Cut a 6×4 parallelogram, slice a right triangle from one end and slide it to the other; write the rule. This is i-Ready Session 1 as published.	ND5 – Identify parallelograms; QMU – Understanding area of a parallelogram

IXL codes are from IXL's published alignment to i-Ready Classroom Mathematics (2024), Grade 6, Unit 1, Lesson 1.

Common Core & Michigan Standards Addressed

Michigan's K-12 Standards for Mathematics adopt the Common Core State Standards; the codes below are identical in both documents.

Grade-Level Standards (Grade 6)

- **6.G.A.1:** Find the area of right triangles, other triangles, special quadrilaterals, and polygons by composing into rectangles or decomposing into triangles and other shapes; apply these techniques to real-world problems. The four-piece split is decomposition; the Enrich board does the parallelogram.
- **6.EE.A.2c:** Evaluate expressions at specific values of their variables, including formulas used in real-world problems. Connect It substitutes into $A = b \times h$; exit ticket problem 3 solves for a missing dimension.

ELA/Literacy Standard Addressed

- **L.6.4:** Determine or clarify the meaning of unknown and multiple-meaning words and phrases, choosing flexibly from a range of strategies. Section A of the classwork worksheet has students infer GROWTH from its letter pattern and context before applying it to multiplication.

Foundational Standards Being Repaired

- **4.NBT.B.5:** Multiply two two-digit numbers using strategies based on place value, illustrated by rectangular arrays and area models. This is the repair standard; Target Dimensions and exit ticket problem 2 assess it.
- **3.MD.C.7:** Relate area to multiplication and addition; use tiling to show that the area of a rectangle with side lengths a and $b + c$ is the sum of $a \times b$ and $a \times c$. The split is 3.MD.C.7c made visible.
- **3.OA.C.7:** Fluently multiply within 100. The daily Facts Sprint and the Engage board.

Standards for Mathematical Practice

- **MP3:** Construct viable arguments and critique the reasoning of others: Discuss It, and the Counter-vs-Splitter skit and vote.
- **MP4:** Model with mathematics: dart throw to rectangle to expression to number.
- **MP7:** Look for and make use of structure: splitting at ten on purpose because it makes the pieces easy.

Pedagogical & Cognitive Research Foundations

Concrete-Representational-Abstract (CRA)

Students build the rectangle on grid paper (concrete), color and label the four pieces (representational), and only then write the expression and the formula (abstract). Witzel, Riccomini, and Schneider (2008) found this sequence particularly effective for secondary students with gaps in prior mathematics, which describes much of this class.

Area Models for Multi-Digit Multiplication

The Common Core progressions documents describe the area model as the bridge between single-digit facts and the standard algorithm, precisely because the model makes place value visible. Fuson and Beckmann (2012) argue that students who can draw and explain the partial-products rectangle understand the algorithm rather than memorize it. Today's lesson introduces no algorithm; the drawing is the method until it is automatic.

Just-in-Time Remediation Over Pull-Out Remediation

Rather than delaying grade-level content to reteach earlier standards, the prerequisite skill is taught inside the grade-level task. TNTP's The Opportunity Myth (2018) found that students who spend their time on grade-appropriate assignments make markedly more progress than peers assigned remedial work, particularly students who start behind.

Retrieval Practice and Distributed Fluency

The Facts Sprint is short, daily, and self-scored. Roediger and Karpicke (2006) showed that brief, repeated retrieval produces more durable learning than longer, less frequent review, which is why fact fluency is handled as a five-minute routine rather than a unit.

Formative Assessment Driving Grouping

Board assignments are made from what the teacher observes during Explore, using a single diagnostic question (“Show me where 10×3 is on your picture”), consistent with Black and Wiliam's (1998) finding that assessment used to adjust instruction in the moment is among the highest-leverage practices available.

Dual Coding and Multimedia Learning

The classwork worksheet's word-then-model design — spelling GROWTH, then drawing the array that shows it — follows dual coding theory (Paivio, 1971) and Mayer's (2009) multimedia-learning principle that pairing verbal and visual representations of the same idea supports comprehension for students who are not yet fluent in either channel alone.

What to Look For as an Observer

- Students using the words “area” and “square units,” not “the answer.”
- When asked, a student can touch the part of the drawing that matches one factor pair in the expression.
- Splits happening at 10 or 20, not at arbitrary lines.
- At the Engage board, students counting rows aloud rather than single squares.
- Darts moving only teacher-to-thrower-to-teacher.
- Students using the classwork worksheet's letter boxes and word bank — rather than abandoning a problem — when spelling or handwriting is the obstacle.
- Exit ticket problem 2 (24×13) completed with a drawing by at least two-thirds of the room. This one result decides whether Day Six runs the parallelogram session as published.

Appendix: Day Five in the Grade 6 Math Lesson Plan Template

Template Field	Day Five Content
Standard(s) & Taxonomy Level	6.G.A.1, 6.EE.A.2c (grade level); 4.NBT.B.5, 3.MD.C.7, 3.OA.C.7 (foundational); L.6.4 (ELA/literacy, classwork worksheet); MP3, MP4, MP7 A/An
Learning Target(s) (SWBAT)	SWBAT find the area of a rectangle by decomposing it into pieces and adding; multiply two two-digit numbers using an area model; and substitute values into $A = b \times h$.
Success Criteria (I Can)	I can find the area of a rectangle by splitting it and adding the pieces. I can multiply two two-digit numbers with a drawing. I can substitute into $A = b \times h$. I can explain where each number in an expression lives on the picture.
Daily Marzano Strategies	Identifying Critical Content; Helping Students Practice Skills, Strategies, and Processes; Helping Students Examine Their Reasoning

Template Field	Day Five Content
Do-Now (5 min)	Facts Sprint: ten single-digit facts, self-scored, plotted on the Human Bar Graph.
Instructional Plan	Try It on a 6×8 grid to establish area = rows $\times$ columns. Target Dimensions: teams generate rectangles with the dart board, draw them, split at ten, and add four partial products. Discuss It / Connect It connects the picture to $(10 + 4)(20 + 3)$ and to $A = b \times h$. Engage/Explore/Enrich rotation ends on aligned IXL skills.
Critical Content Debrief (5 min)	Can students locate each partial product on the drawing, and can they complete 24×13 with the split?
Daily Formative [Ungraded] / Summative [Graded] Assessments	Formative: sprint score, teacher observation during Explore (the 10×3 question), journal, classwork worksheet. Summative: none today; the exit ticket is diagnostic and sets Day Six grouping.

Planning to Close the Achievement Gap Using Data

Tier 1 Supports	Tier 2 Supports	Tier 3 Supports
Whole-class model round on the document camera before teams throw; grid paper for every problem; numbers kept under 25 by design; letter-boxed name field and word bank on the classwork worksheet for every student.	Engage board with the teacher: arrays counted by rows, single-split rectangles, IXL CQZ; sentence frames for Discuss It ("This piece is $___ \times ___$ because $___$ ").	Names from the Engage board go to this week's 1-on-1 tutoring list for skip-counting and arrays; pre-gridded rectangles with the tens line already drawn.

Data Source: Days One-Four formative observations; Facts Sprint; Day Five exit ticket; classwork worksheet.

Explore More: Resources for Observers

For guests who would like to look deeper into the curriculum and standards behind today's lesson.

About Today's Curriculum

IXL Alignment to i-Ready Classroom Mathematics, Grade 6

The published skill-by-skill alignment used to choose today's IXL board tasks (Unit 1, Lesson 1).

[View the skill plan](#)

Michigan K-12 Math Standards

The Michigan Department of Education's full K-12 Standards for Mathematics document.

[View Michigan DOE PDF](#)

Today's Classwork Worksheet — Multiplication Means Growth

The Guess the Word / Meet the Multiplication
Family worksheet referenced throughout this guide.

[View the worksheet](#)

For Deeper Research & Standards

Standards for Mathematical Practice

The official Common Core description of MP3,
MP4, MP7, and the other five practices.

[View on corestandards.org](http://corestandards.org)

Inside the Black Box (Summary)

A widely cited summary of Black and Wiliam's
formative-assessment research, the basis for
today's in-the-moment grouping.

[Read the summary](#)

References

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